Matthew Morgan

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EDUCATION

Iowa State University Ames, Iowa — Expected Graduation: December 2026
Bachelor of Science, **Software Engineering** — GPA 4.00/4.00

WORK EXPERIENCE

Ecolab, Naperville, IL

Innovation Engineer — August 2024 – Present

- Rapidly developed revenue generating prototypes by leveraging our data to solve customer problems.
- Developed a novel approach to leveraging free text data across multiple data sources for customer success
- Served as a **flexible technical contributor**, working on high-impact projects wherever additional resources or expertise was most needed.
- Continued to develop AI prototypes, internal demos, and customer solutions for top level executives.

Artificial Intelligence Intern — May 2024– August 2024

- Developed Generative AI prototype to live-demo for the largest Ecolab customer.
- Built a product that identifies **$X millions** of missing Total Value Delivered, where our customers are not utilizing our products and services to the proper extent, which leads to customer safety risks.
- Created a **rapid prototyping** AI environment so we could make constant iterations of the prototype as we worked towards the final product to be presented to the Board.
- Shadowed and **mentored** under the **Director of Artificial Intelligence**.
- Collaborated with a team of 3 to successfully deliver an AI Innovation for **Ecolab's Board of Directors**.

Bear Paddle Swim School, Aurora, IL

Deck Specialist — April 2022– September 2023

- Organized and led team meetings before and after each shift to improve each teachers' lessons.
- Coached new or struggling teachers to improve quality of lessons and customer satisfaction.
- Communicated with customers to gain feedback and enhance the overall quality of our services.

Technical Projects

Robotic Arm with Computer Vision Control | Robotics/Computer Vision — March 2025– Present

- Built a **closed-loop robotic arm** system following the Open-Source Arctos v2.9.7 platform
- Developed a React-based web interface to visualize camera input and control the robot wirelessly.
- Developed a **vision pipeline** in Python with **OpenCV** to track objects for autonomous manipulation.

Camera Based Golf Simulator | Computer Vision — October 2023 – April 2025

- Employed **Python** and **OpenCV** to apply multiple filters in order to identify where a golf ball is located in each frame of a slow-motion video
- Used the ball's location data to calculate the trajectory and speed of a golf ball directly after impact.

LEADERSHIP

- State Storm Drumline Section Leader — September 2024 - Present
- Several Leadership Seminars as a Marching Band Section Leader — September 2021 - Present
- Foundations of Leadership Development and Learning Course — August 2023 - December 2023

SKILLS

Google Suite, Python, Pandas, NumPy, OpenCV, TensorFlow, Scikit-Learn, Flask, Gen-AI, Java, C, C++, HTML, CSS, JavaScript, React.JS, TypeScript, Selenium, MySQL, Webscraping, LangChain, LangGraph, NodeJS.